



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies

- Data collection from SpaceX API and Wikipedia.
- Data cleaning and feature engineering.
- Exploratory analysis using SQL and maps.
- Interactive dashboard development.
- Machine learning model training and testing

- Summary of all results

- KSC LC-39A had the best recovery performance.
- Payload range 2,000–6,000 kg showed better outcomes
- All models reached 83.33% accuracy.
- Logistic Regression was the most suitable model.
- SpaceX showed improved reliability over time.

Introduction

- Project background and context
 - SpaceX reduces launch costs by reusing boosters.
 - A successful booster landing improves cost efficiency and mission sustainability.
 - Landing outcomes vary based on
 - payload,
 - orbit,
 - launch site, and
 - booster version.
 - Historical data helps explain these patterns.
- Problems you want to find answers
 - What factors most affect booster landing success?
 - How Falcon 9 landing success improved?
 - Can ML predict landing outcomes accurately?
 - How can these results improve launch decisions

Section 1

Methodology

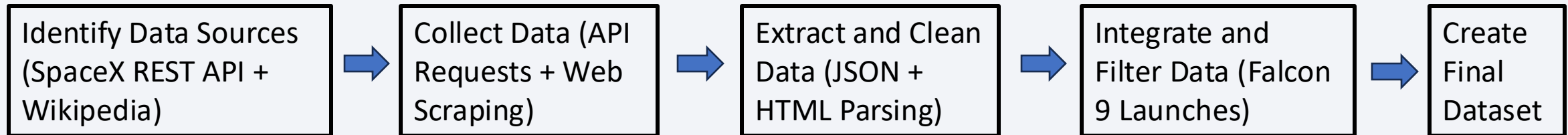
Methodology

Executive Summary

- Data collection methodology:
 - The data was collected from the SpaceX API and Wikipedia to create a complete Falcon 9 launch dataset.
- Perform data wrangling
 - The data was cleaned and transformed by handling missing values and organizing the features into a structured dataset ready for further analysis.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Classification models were trained on split data, refined with GridSearchCV, and assessed using accuracy and a confusion matrix to show the strongest model.

Data Collection

- Collected Falcon 9 launch data from the SpaceX API and Wikipedia.
- Used API requests to extract mission data in JSON format.
- Used web scraping to collect added launch records from HTML pages.
- Combined and filtered both sources into one complete dataset.



Data Collection – SpaceX API

<https://github.com/omarfaruqe/DS-Capstone/blob/main/jupyter-labs-spacex-data-collection-api-v2.ipynb>



Data Collection - Scraping

<https://github.com/omarfaruqe/DS-Capstone/blob/main/jupyter-labs-webscraping.ipynb>

Open Wikipedia Page



Locate Falcon 9 Launch Table



Parse HTML with BeautifulSoup



Extract Table Headers and Rows



Clean and Normalize Text



Build Pandas DataFrame



Save Structured Dataset

Data Wrangling

- Loaded the SpaceX dataset into a pandas DataFrame for structured analysis.
- Reviewed missing values and checked data types to understand overall data quality.
- Explored categorical variables such as **LaunchSite**, **Orbit**, and **Outcome** to identify patterns.
- Counted launches by site and calculated the occurrence of each orbit type.
- Examined landing outcomes and grouped unsuccessful cases into a `bad_outcomes` set.
- Binary target column, **Class**, where 1 and 0 shows a successful and unsuccessful landing.
- Exported the cleaned and labeled dataset for use in later modeling steps.

Reference Notebook

- <https://github.com/omarfaruqe/DS-Capstone/blob/main/labs-jupyter-spacex-Data-wrangling-v2.ipynb>

EDA with Data Visualization

Charts used

- **Launch Site vs Flight Number scatter plot** — shows success trends by site over time.
- **Launch Site vs Payload Mass scatter plot** — shows how payload size affects success by site.
- **Orbit Success-Rate Bar Chart** — compares average success across orbit types.
- **Flight Number vs Orbit scatter plot** — shows whether success improves with more flights.
- **Payload Mass vs Orbit scatter plot** — shows how payload and orbit relate to outcomes.
- **Yearly Success Trend Line chart** — shows how success rate changes over time.

Why these charts

- **Scatter plots** were chosen when the goal was to compare two variables and show class labels at the same time
 - **Bar charts** were used when the goal was to compare success rates across categories
 - **Line charts** were used for the yearly trend because they are the clearest way to show change over time.
-
- <https://github.com/omarfaruqe/DS-Capstone/blob/main/jupyter-labs-eda-dataviz-v2.ipynb>

EDA with SQL

SQL Queries Performed

- Distinct launch sites — `SELECT DISTINCT Launch_Site FROM SPACEXTBL;`
- First 5 launches — `SELECT * FROM SPACEXTBL LIMIT 5;`
- Total payload mass — `SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTBL;`
- Average payload mass — `SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTBL;`
- Earliest launch date — `SELECT MIN(Date) FROM SPACEXTBL;`
- F9 FT boosters — `SELECT Booster_Version FROM SPACEXTBL WHERE Booster_Version LIKE 'F9 FT%';`
- Mission outcome counts — `SELECT Mission_Outcome, COUNT(*) FROM SPACEXTBL GROUP BY Mission_Outcome;`
- Falcon 9 Block 5 boosters — `SELECT Booster_Version FROM SPACEXTBL WHERE Booster_Version LIKE 'F9 B5%';`
- Failed drone ship landing by month — `SELECT substr(Date, 6, 2) AS Month, Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTBL WHERE Landing_Outcome = 'Failure (drone ship)';`
- Landing outcome counts — `SELECT Landing_Outcome, COUNT(*) AS Count FROM SPACEXTBL GROUP BY Landing_Outcome;`
- https://github.com/omarfaruqe/DS-Capstone/blob/main/jupyter-labs-eda-sql-coursera_sqlite.ipynb

Build an Interactive Map with Folium

- Created a Folium base map to display SpaceX launch sites and nearby geographic features.
- Added circle markers and text labels to show each launch site clearly.
- Used clustered green and red markers to show successful versus failed launches without cluttering the map.
- Added mouse-position coordinates to help find nearby reference points accurately.
- Added distance markers and polylines to show how close launch sites are to features.

<https://github.com/omarfaruqe/DS-Capstone/blob/main/lab-jupyter-launch-site-location-v2.ipynb>

Build a Dashboard with Plotly Dash

- **Pie chart:** to show launch success totals by site, or success vs. failure for a selected site.
 - This helps compare site performance quickly.
- **Scatter plot:** Payload mass vs. landing outcome, colored by booster version category.
 - This helps reveal how payload size and booster type relate to mission success.
- **Launch site dropdown:** Users can switch between all sites and a specific launch site.
 - This makes the dashboard interactive and supports site-level analysis.
- **Payload range slider:** Users can filter launches by payload mass.
 - This helps focus the analysis on specific payload intervals and see how success patterns change.
- https://github.com/omarfaruqe/DS-Capstone/blob/main/spacex_dash_app.py

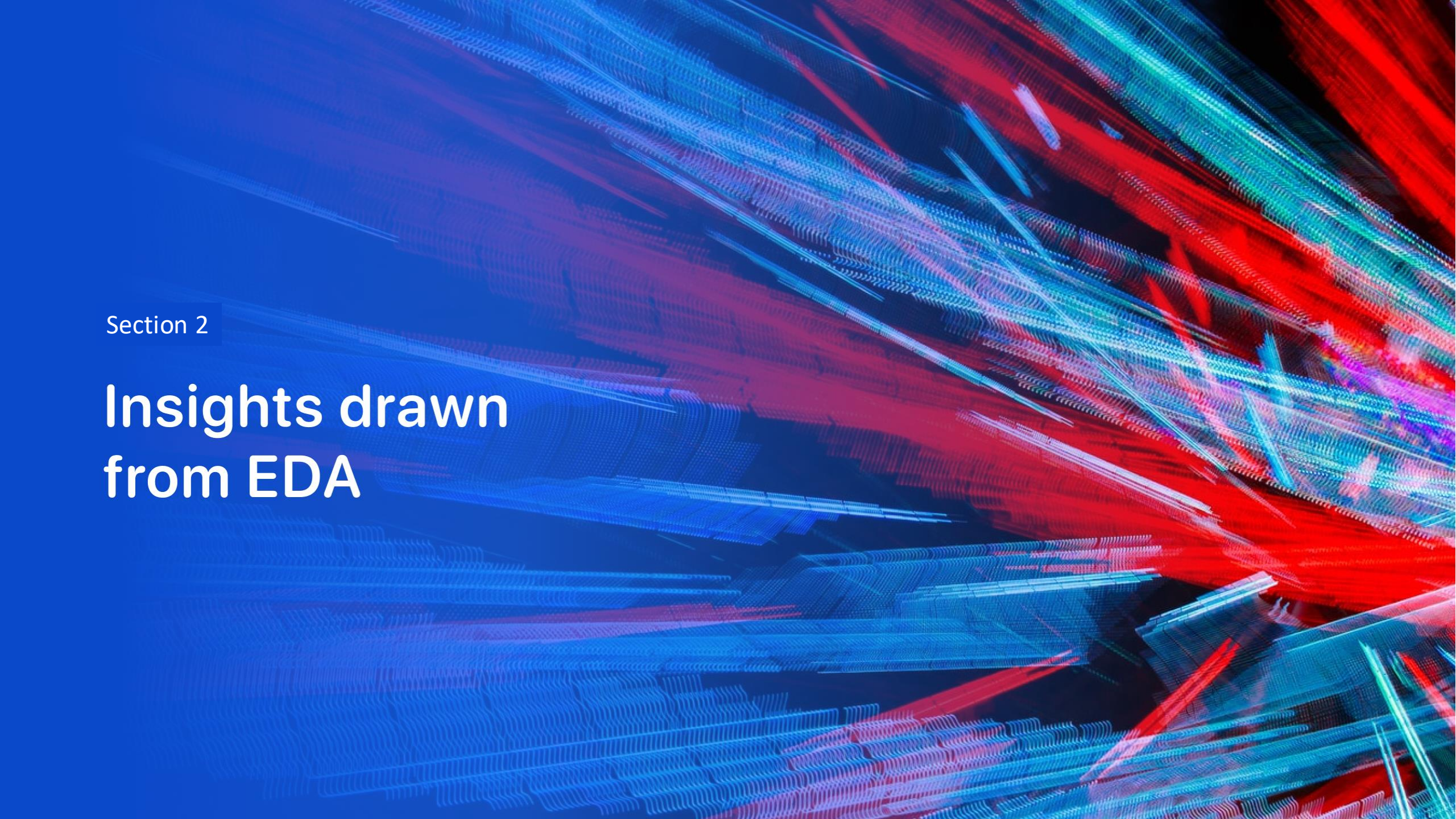
Predictive Analysis (Classification)

- Standardized features, then split the data into 80% training and 20% testing sets.
- Built and tuned Logistic Regression, SVM, Decision Tree, and KNN using GridSearchCV with 10-fold cross-validation.
- Evaluated models using validation accuracy, test accuracy, and confusion matrices.
- All models achieved 83.33% test accuracy; Logistic Regression was selected as the best model due to its robust performance and efficiency.

<https://github.com/omarfaruqe/DS-Capstone/blob/main/SpaceX-Machine-Learning-Prediction-Part-5-v1.ipynb>

Results

- Exploratory data analysis results
 - Collect launch data from SpaceX datasets and launch records.
 - Map launch sites and classify success or failure outcomes.
 - Measure proximity to coastline, railways, highways, and populated areas.
 - Run SQL analysis to find landing trends, payload averages, and milestone dates.
 - Summarize the key EDA insights for use in a presentation slide.
- Predictive analysis results
 - Four models were tested, and all reached 83.33% accuracy in predicting outcomes.
 - Logistic Regression was chosen as the best overall model because it combined strong accuracy with simplicity and interpretability.
 - The SpaceX results show that landing success can be predicted reliably from mission data.
 - The tumor model showed higher recall with Random Forest, helping reduce dangerous false negatives in diagnosis.
 - Robust performance under noisy data suggests good reliability in real-world use.

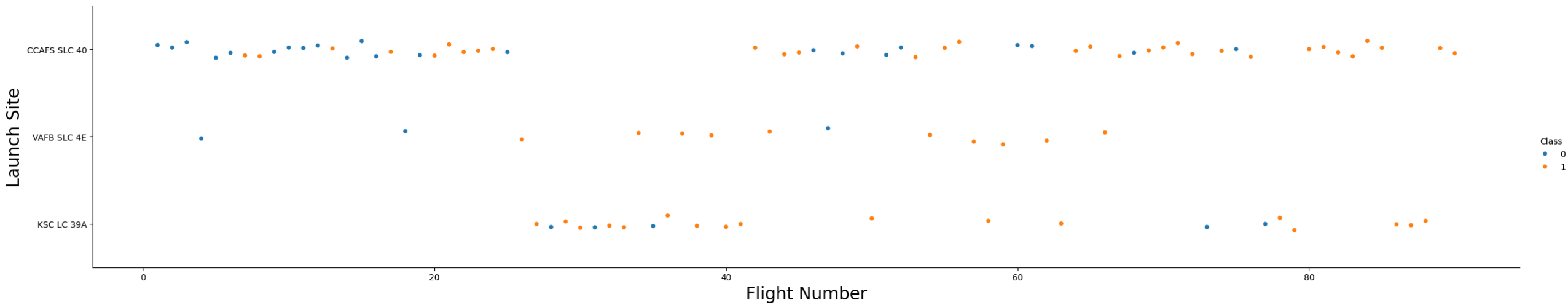
The background of the slide is an abstract composition. It features a dark blue field on the left side, which transitions into a complex pattern of diagonal streaks in shades of blue, red, and cyan on the right. These streaks have a textured, almost woven appearance. Overlaid on this pattern is a faint, light blue grid that recedes into the distance, creating a sense of depth and perspective.

Section 2

Insights drawn from EDA

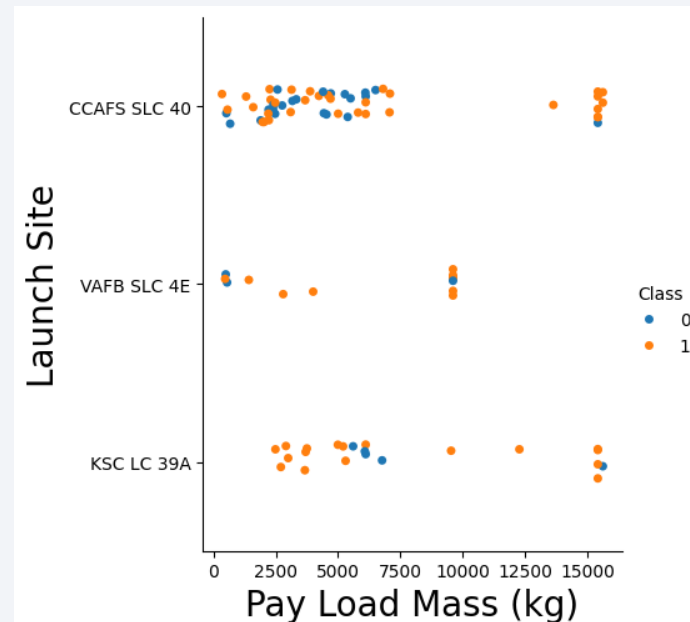
Flight Number vs. Launch Site

- CCAFS SLC 40 has the highest number of launches in the dataset.
- Orange dots indicate successful landings; blue dots indicate failures.
- More orange dots appear at higher flight numbers, showing improvement over time.
- KSLC 39A was introduced later and shows a notably high success rate.
- The pattern suggests launch success increased with operational experience.



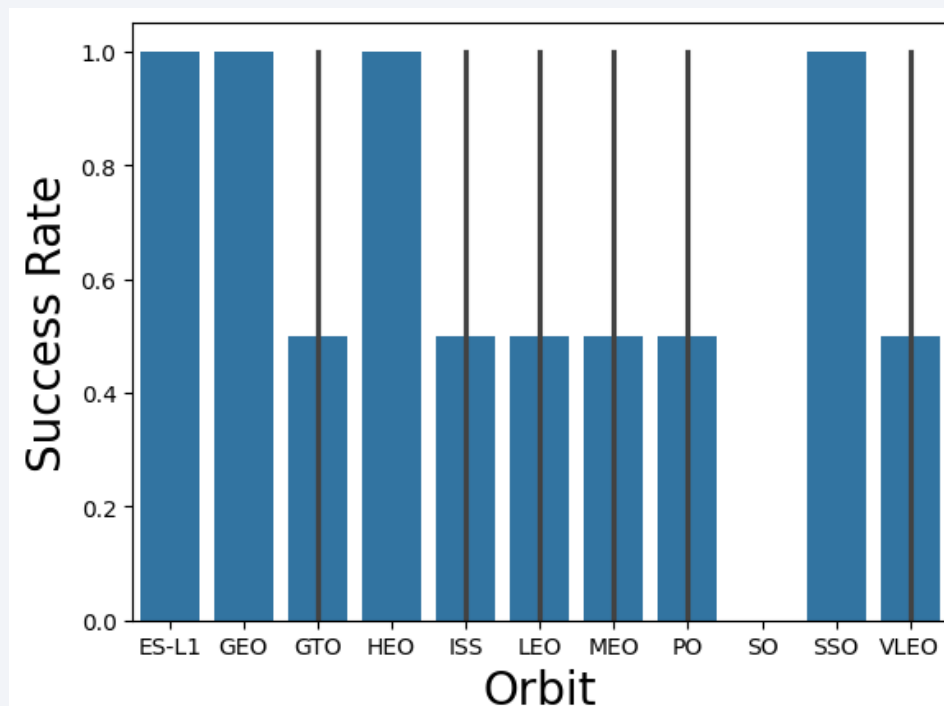
Payload vs. Launch Site

- Most launches across sites are successful, shown by the higher number of orange points.
- Payload mass varies widely, especially at CCAFS SLC 40.
- CCAFS SLC 40 supports most launches and performs well with payloads above 8,000 kg.
- Lighter payloads at CCAFS SLC 40 show mixed results.
- VAFB SLC 4E does not include launches above 10,000 kg in this dataset.
- Launch success appears to depend on more than payload mass alone.



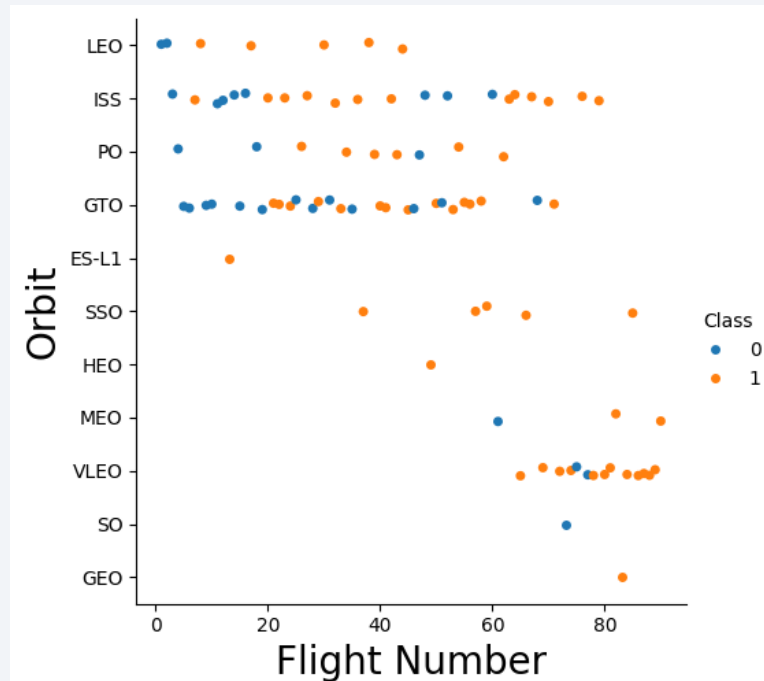
Success Rate vs. Orbit Type

- The bar chart shows clear differences in landing success rate across orbit types.
- ES-L1, GEO, HEO, and SSO reach a 100% success rate in the dataset.
- GTO missions have a success rate of 50%.
- Success variability differs by orbit, showing that some mission types are more challenging than others.
- Orbit type is a strong factor in recovery performance.



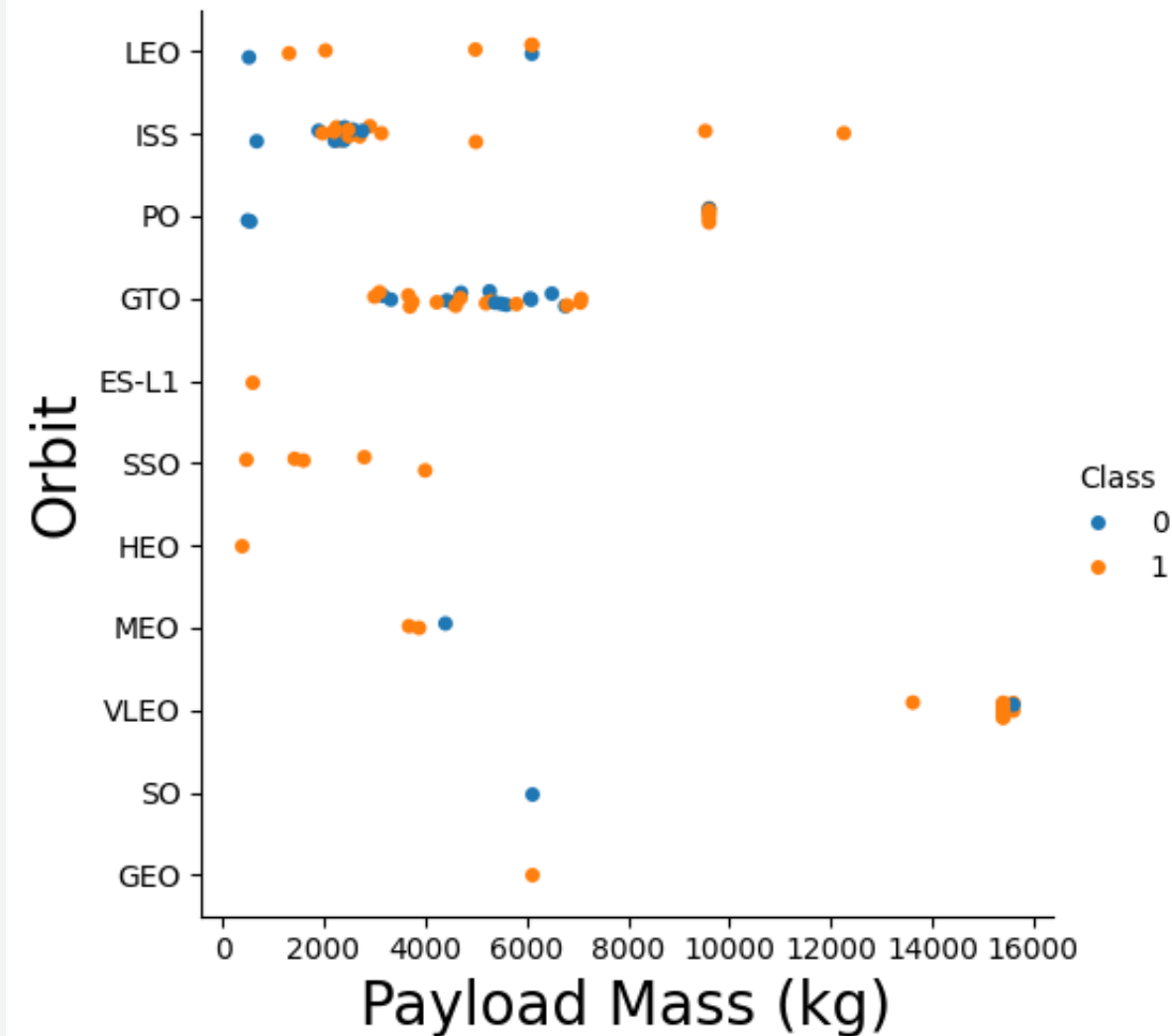
Flight Number vs. Orbit Type

- Most launches across orbit types are successful, shown by the larger number of orange points.
- Orbit categories such as SSO, HEO, GEO, and GEO1 show mostly successful outcomes.
- Earlier flights have a more mixed pattern of success and failure.
- Later flight numbers show a higher concentration of successful missions.
- The overall pattern shows improved mission performance over time.



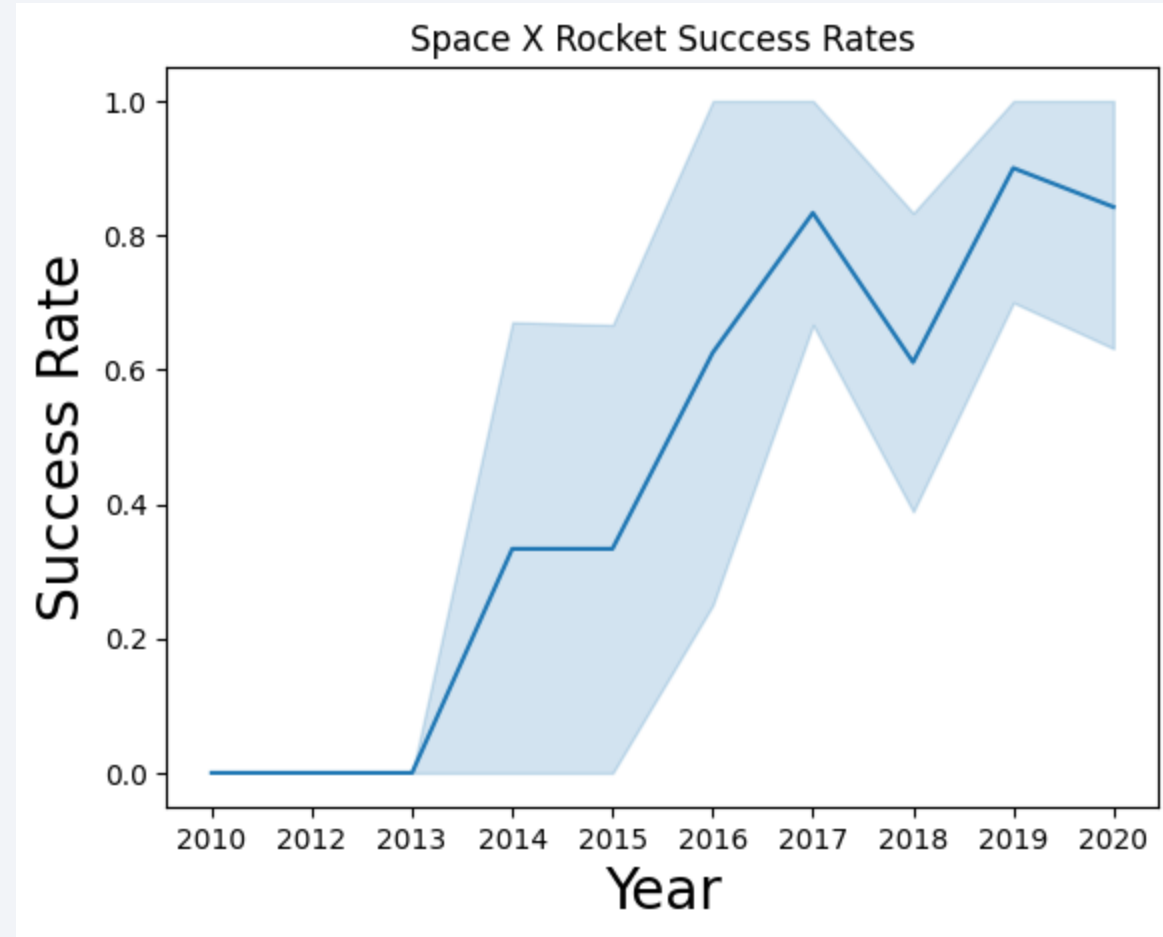
Payload vs. Orbit Type

- GTO is mostly successful, with payloads concentrated around 3,000 to 6,500 kg.
- GEO is almost entirely successful and generally involves heavier payloads, often above 6,000 kg.
- SSO missions are mostly successful and usually carry lighter payloads below about 4,000 kg.
- LEO and ISS show the widest payload range and a more mixed success pattern.
- MEO, HEO, and ES-L1 are sparse and show fewer missions with mixed results.
- Very heavy payloads above 10 to 12 tons are uncommon and appear mainly in LEO, VLEO, and GEO-related missions.



Launch Success Yearly Trend

- The scatter plot links payload mass and orbit type to landing success.
- Very heavy VLEO payloads, around 15,000 kg, show a notably high success rate.
- This VLEO pattern is consistent with optimized Starlink deployment missions.
- GTO missions typically carry lighter payloads, around 4,000 to 8,000 kg, but still show mixed outcomes.
- The chart suggests that orbit energy requirements can matter more than payload mass for booster recovery.



All Launch Site Names

- The query used was `SELECT DISTINCT launchsite FROM SPACEXTABLE.`
- `DISTINCT` helps remove duplicate launch site names.
- The dataset has four unique launch site entries.
- They are
 - CCAFS LC-40
 - VAFB SLC-4E
 - KSC LC-39A, and
 - CCAFS SLC-40.

Launch Site Names Begin with 'CCA'

- The SQL query used was

```
SELECT * FROM SPACEXTABLE WHERE launchsite LIKE 'CCA%' LIMIT 5.
```

- This returns the first 5 records where the launch site name begins with **CCA**.
- The query used LIKE 'CCA%' to filter launch sites by prefix.
- The matching launch site shown in the sample records is **CCAFS LC-40**

Total Payload Mass

- The SQL query used was

```
SELECT SUM(PAYLOADMASSKG) FROM SPACEXTABLE WHERE customer LIKE 'NASA%'
```

- This adds up the payload mass for all missions associated with NASA customers.
- The result is **45,596 kg**.
- This total shows the combined payload carried on NASA-related launches in the dataset.
- This query is useful because it measures how much cargo SpaceX delivered for NASA missions overall.
- The SUM function calculates the total payload mass, and the LIKE 'NASA%' filter selects the NASA records

Average Payload Mass by F9 v1.1

- The SQL query used was

```
SELECT AVG(PAYLOADMASSKG) FROM SPACEXTABLE WHERE boosterversion  
LIKE 'F9 v1.1%'
```

- This calculates the average payload mass for launches using the F9 v1.1 booster version.
- The result is **2,928.4 kg**.
- This means the F9 v1.1 booster carried an average payload of about 2.9 metric tons across the missions in the dataset.
- This query uses the AVG function to find the mean payload mass, and the LIKE filter selects only records for the F9 v1.1 booster version. It gives a quick summary of the typical payload size for that booster.

First Successful Ground Landing Date

- The first successful ground pad landing date is **2015-12-22**. The query used is:

```
SELECT MIN(Date) FROM SPACEXTABLE WHERE Landing_Outcome = 'Success  
(ground pad) '
```

- This query finds the earliest date among records where the landing outcome was a successful ground pad landing.

Successful Drone Ship Landing with Payload between 4000 and 6000

- Query used:

```
SELECT boosterversion FROM SPACEXTABLE WHERE missionoutcome LIKE  
'Success' AND payloadmasskg BETWEEN 4000 AND 6000 AND  
landingoutcome = 'Success drone ship'
```

- Matching boosters:

- **F9 FT B1022.**
- **F9 FT B1026.**
- **F9 FT B1021.2.**
- **F9 FT B1031.2.**

- This result shows the booster versions that met two important conditions at the same time: successful drone ship landing and medium-heavy payload delivery.
- It helps highlight which Falcon 9 boosters performed well in that payload range.

Total Number of Successful and Failure Mission Outcomes

- The total mission outcomes in the notebook are **98 successful** and **1 failure**.
- The SQL used in the notebook groups the data by missionoutcome and counts each category:

```
SELECT missionoutcome, COUNT(*) as total_number FROM SPACEXTBL  
GROUP BY missionoutcome;
```

- This query shows how many missions ended in each outcome category.
- It helps summarize the overall launch performance of SpaceX in the dataset.

Boosters Carried Maximum Payload

```
SELECT boosterversion FROM SPACEXTABLE WHERE payloadmasskg = (SELECT  
MAX(payloadmasskg) FROM SPACEXTABLE)
```

- The query returns all boosters associated with the maximum payload value.
- Matching boosters include **F9 B5 B1048.4, F9 B5 B1049.4, F9 B5 B1051.3, and F9 B5 B1056.4.**
- The full list also includes **B1048.5, B1051.4, B1049.5, B1060.2, B1058.3, B1051.6, B1060.3, and B1049.7.**
- This result shows that several Falcon 9 Block 5 boosters reached the same top payload level.
- It highlights the booster versions used for the heaviest missions in the data

2015 Launch Records

```
SELECT substr(Date, 6, 2) AS Month, Landing_Outcome,  
Booster_Version, Launch_Site FROM SPACEXTABLE WHERE substr(Date,  
1, 4) = '2015' AND Landing_Outcome = 'Failure (drone ship)' ORDER  
BY Month;
```

- The failed drone ship landing records appear in **January** and **April 2015**.
- Booster versions include **F9 v1.1 B1012** and **F9 v1.1 B1015**.
- The launch site shown for these records is **CCAFS LC-40**.
- This query is useful for reviewing early drone ship recovery attempts in 2015.
- It shows which boosters failed, when the failures happened, and where the launches took place.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
SELECT Landing_Outcome, COUNT(*) AS Count FROM SPACEXTABLE WHERE  
Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome  
ORDER BY Count DESC;
```

- No attempt: 10.
 - Success (drone ship): 5.
 - Failure (drone ship): 5.
 - Success (ground pad): 3.
 - Controlled ocean: 3.
 - Uncontrolled ocean: 2.
 - Failure (parachute): 2.
 - Precluded (drone ship): 1.
- This query helps compare how often each landing outcome occurred during the early Falcon 9 launch period.
 - It shows that mission outcomes were diverse, with failures and successes both appearing across drone ship, ground pad, and ocean recovery categories.

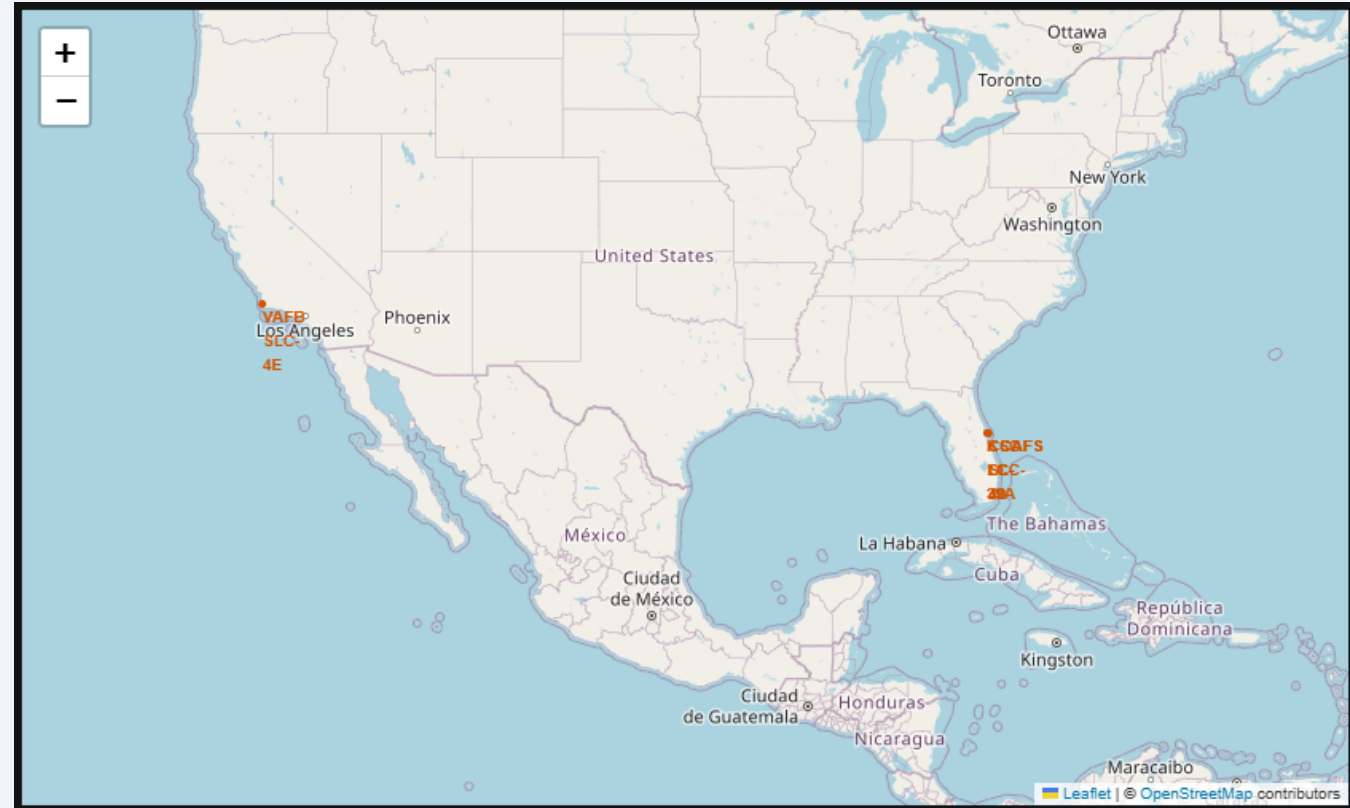
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a solid blue background on the left and a satellite photograph of Earth on the right. The Earth's surface is dark, with numerous bright yellow and orange lights representing cities and urban areas. The horizon of the Earth is visible as a curved line separating the dark surface from the deep blue of space.

Section 3

Launch Sites Proximities Analysis

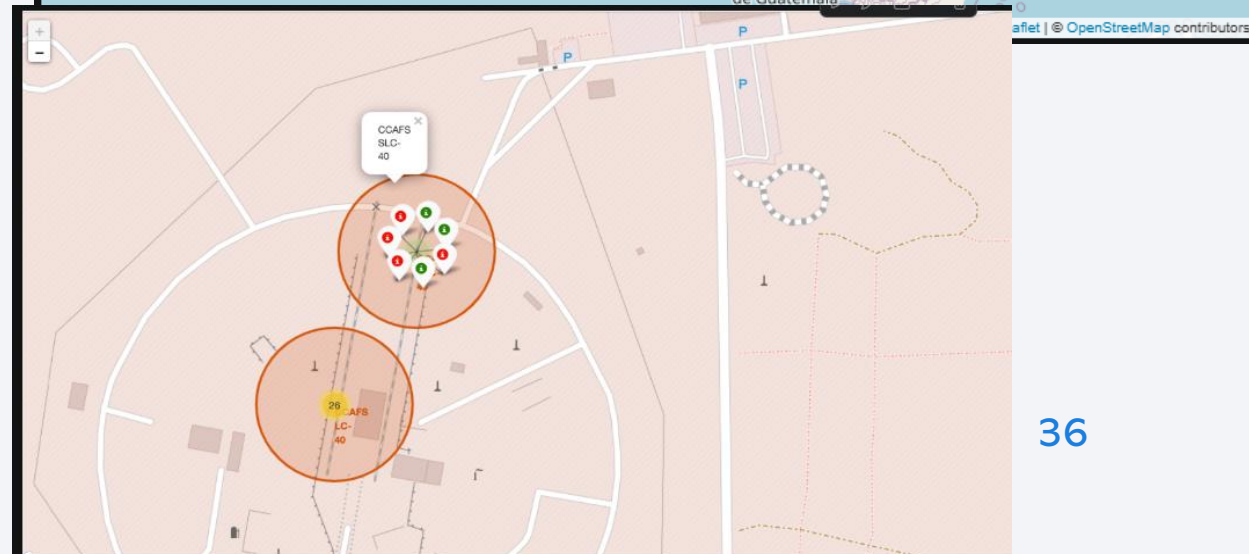
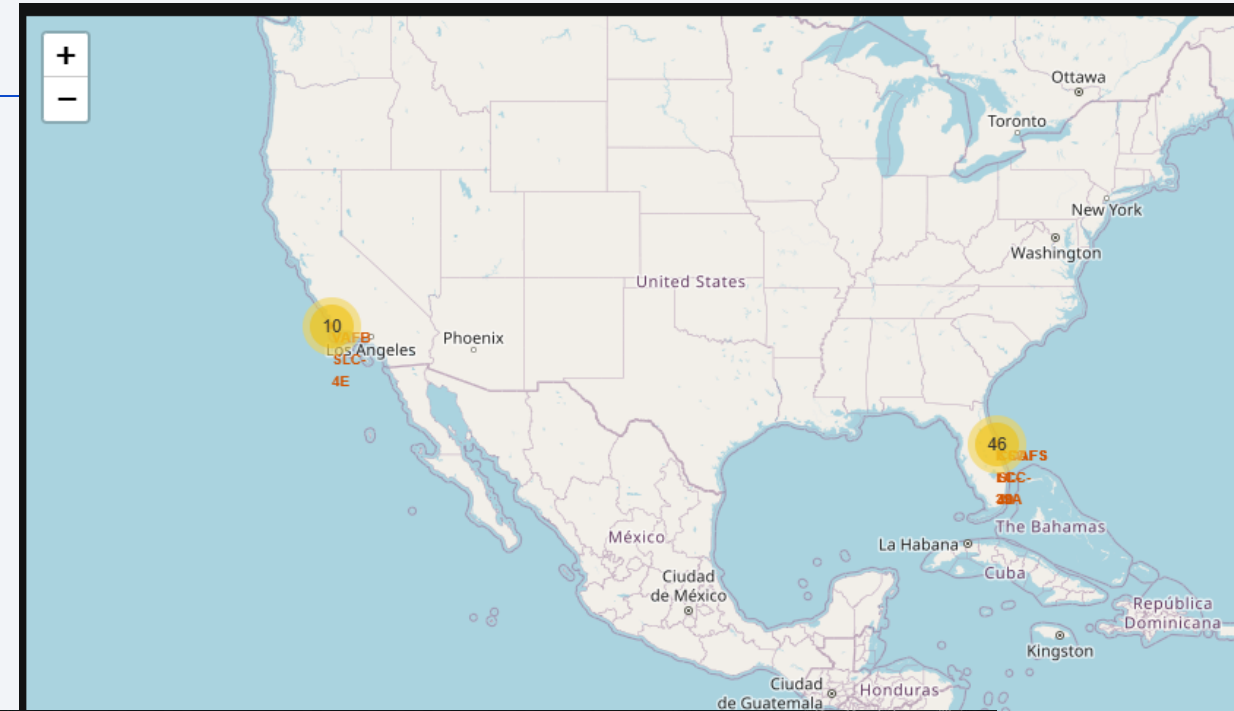
Geographical Analysis of SpaceX Launch Sites

- The Folium map marks four SpaceX launch sites in Florida and California.
- Florida contains three nearby east coast sites: **CCAFS LC-40, CCAFS SLC-40, and KSC LC-39A.**
- **VAFB SLC-4E** is the western launch site in California.
- All mapped launch sites are close to the coast, improving launch safety over open water.
- Green and red clustered markers show successful and unsuccessful launches by site.
- The notebook for this lab does not include a Texas launch-site marker.



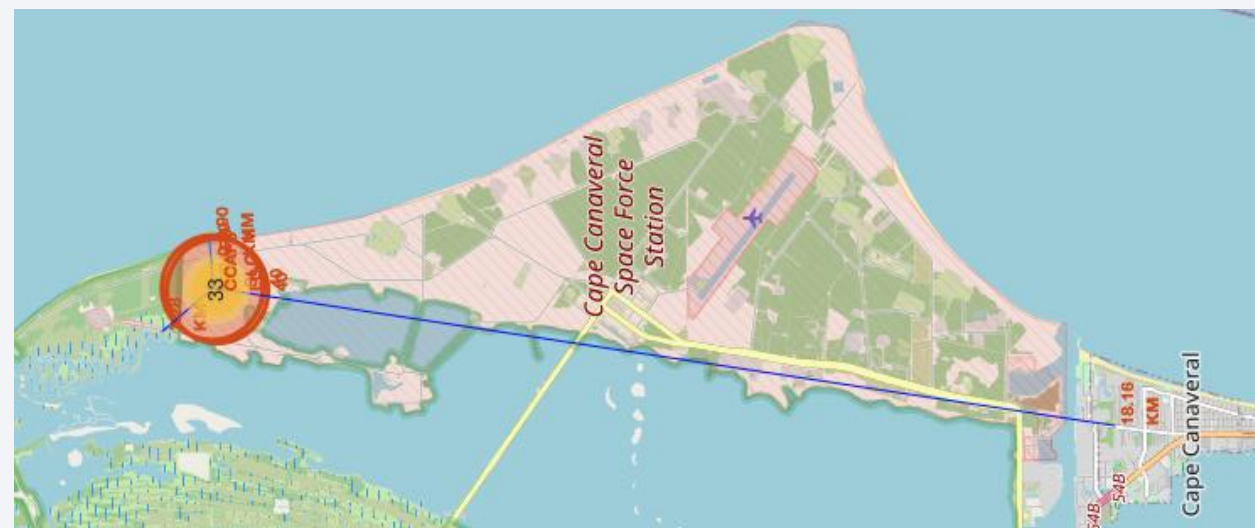
Landing Success Clusters

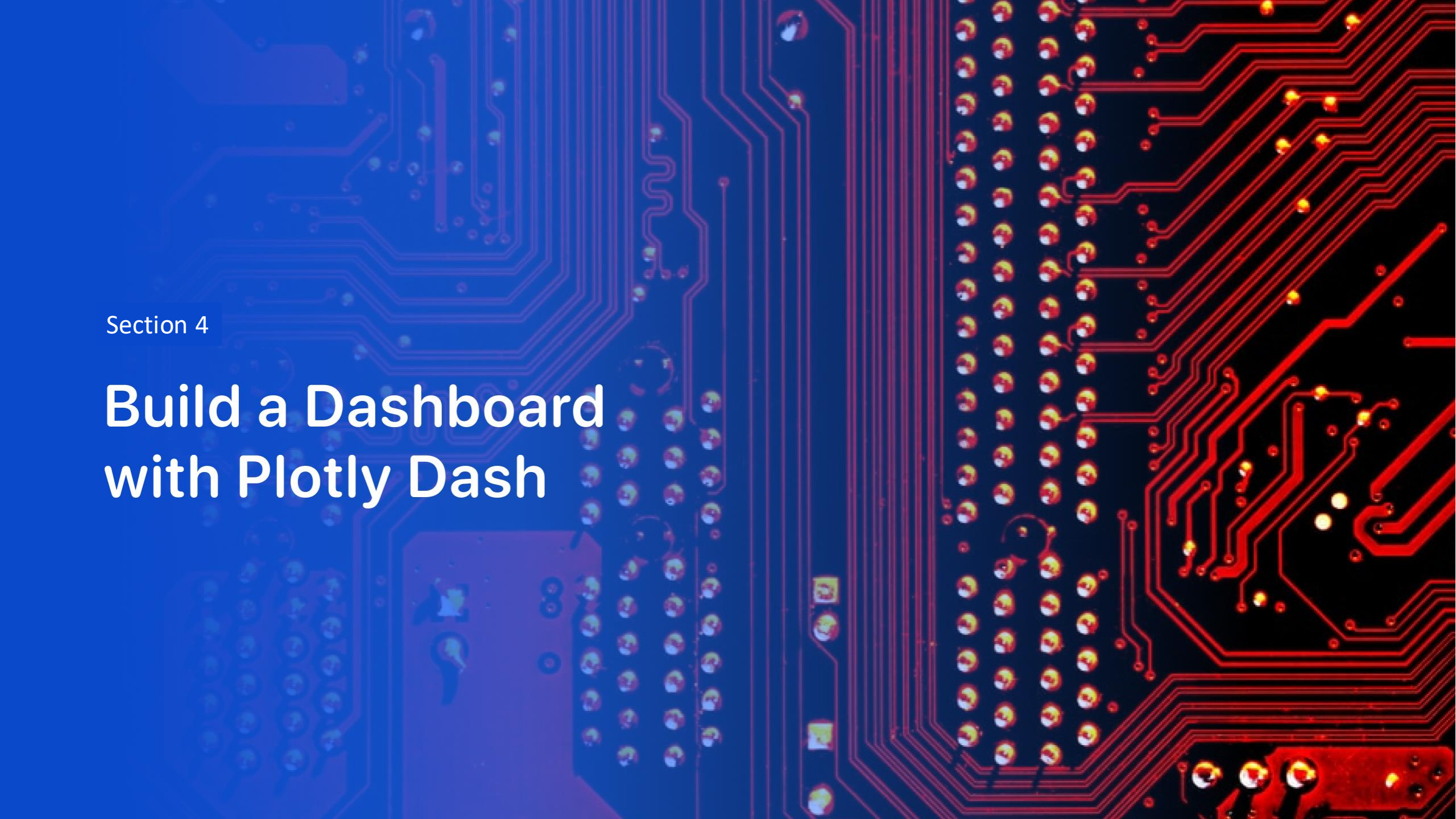
- Site labels show the major launch locations included in the notebook: **CCAFS LC-40**, **CCAFS SLC-40**, **KSC LC-39A**, and **VAFB SLC-4E**.
- Green markers show successful launches; red markers show failed or unsuccessful outcomes.
- Marker clusters are used because many launch records share the same latitude and longitude.
- Florida shows the highest concentration of launches, especially around **CCAFS SLC-40**.
- **KSC LC-39A** stands out visually for its high share of successful outcomes.



Launch Site Proximity (Safety & Logistics)

- The map zooms in on **CCAFS SLC-40** and measures distance to nearby features.
- A PolyLine and map label show the calculated distance visually.
- The notebook example shows the nearest coastline at about **0.86 km** from the launch site.
- The site is close to the coast for safety, but still accessible for logistics.
- This supports the idea that launch sites balance isolation with transport connectivity.



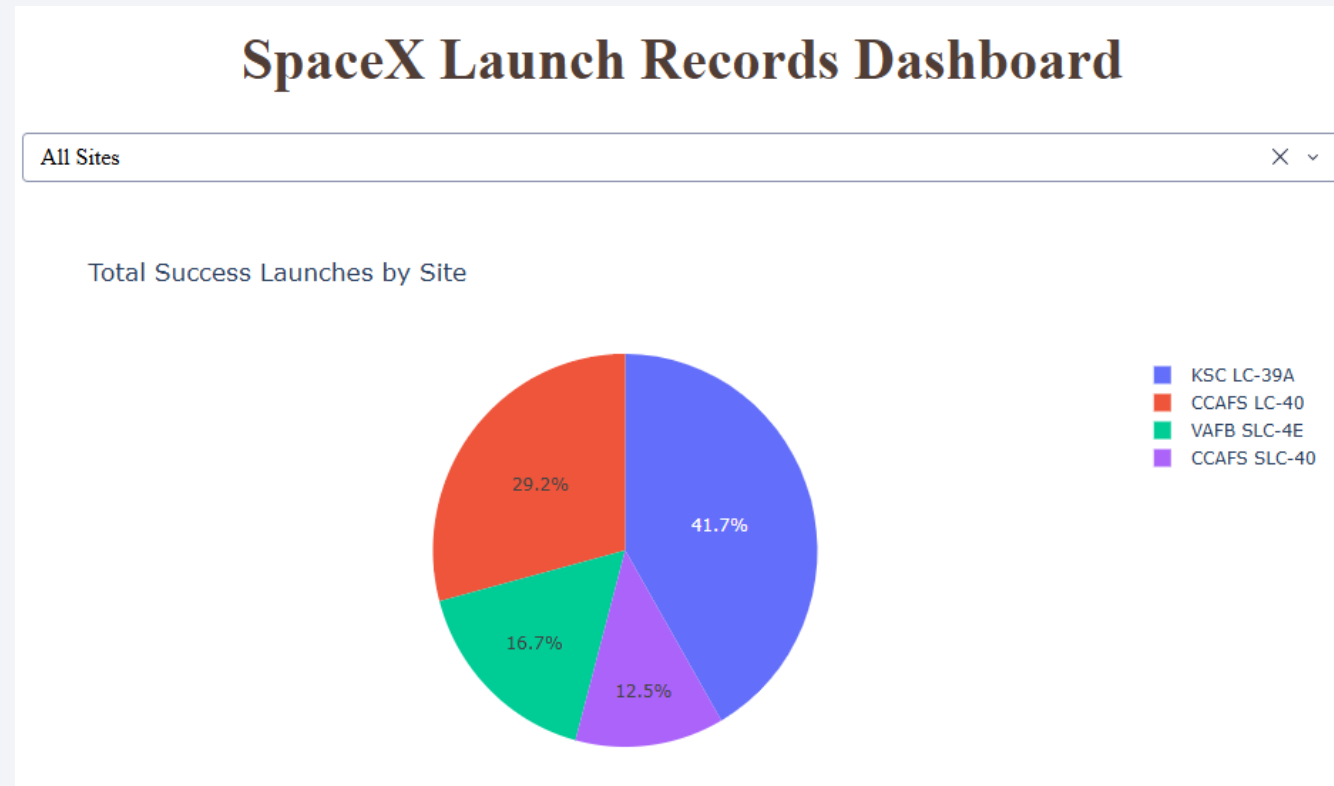


Section 4

Build a Dashboard with Plotly Dash

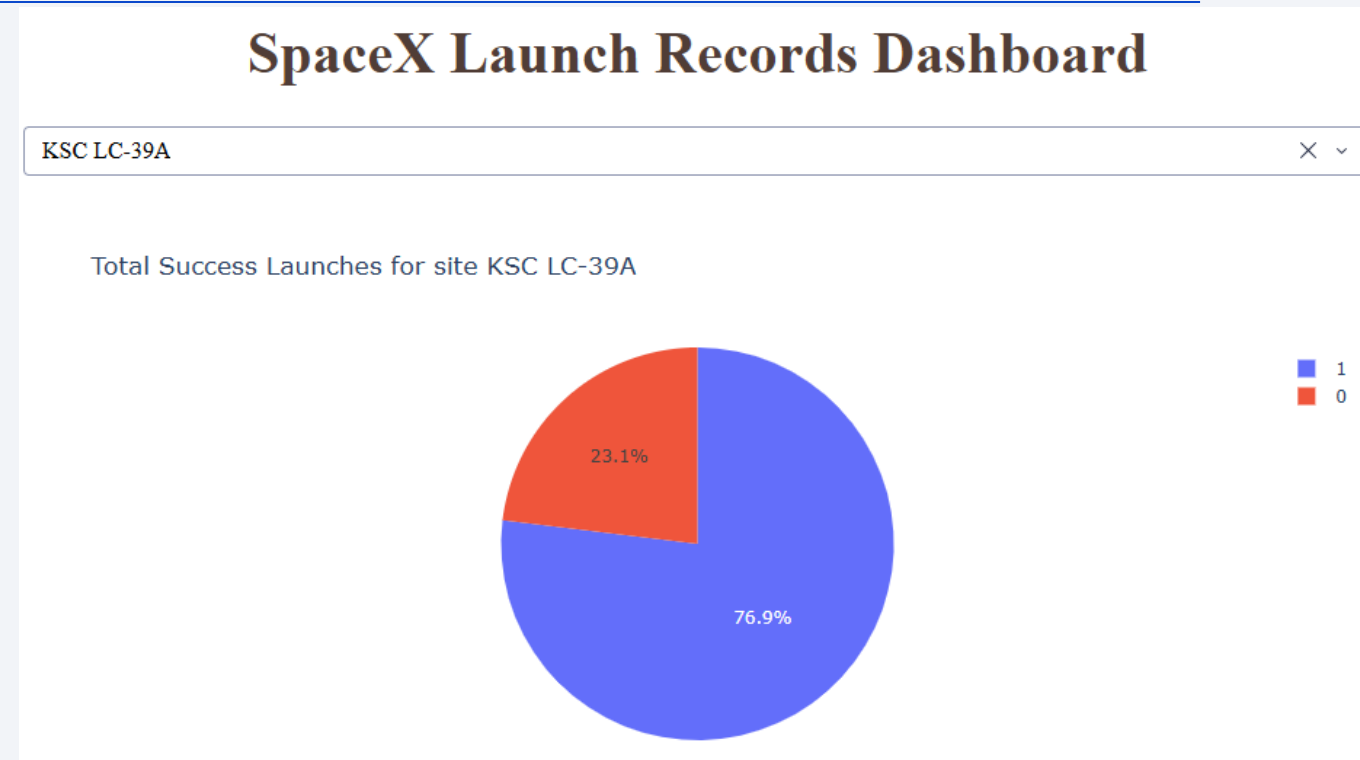
Successful Launches by SpaceX LaunchSite

- Title: **KSC LC-39A** has the largest share of successful launches at about **41.7%**.
- **CCAFS LC-40** is second with about **29.2%**.
- **VAFB SLC-4E** contributes about **16.7%**.
- **CCAFS SLC-40** has the smallest share at about **12.5%**.
- The chart shows that Florida launch sites dominate overall launch success in the dataset.



KSC LC-39A Launch Success Ratio

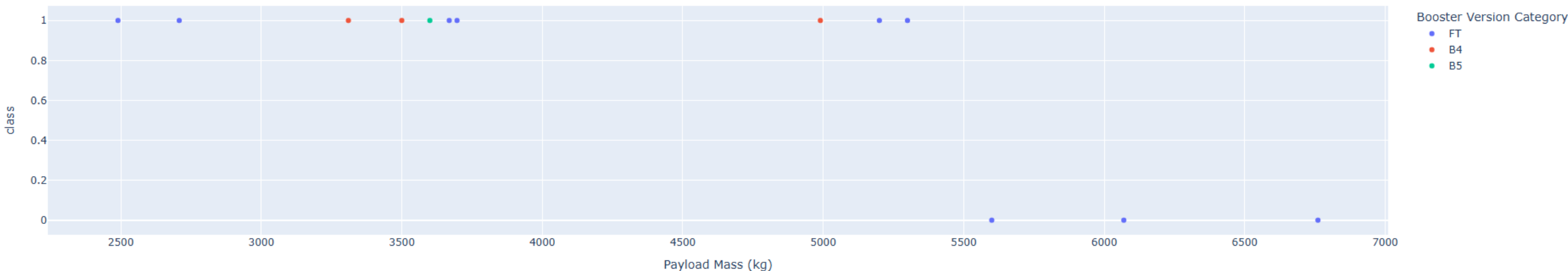
- **KSC LC-39A** has the highest launch success rate in the dataset.
- About **76.9%** of launches are successful and **23.1%** are unsuccessful.
- Successful launches outnumber failures by about **3.3 to 1**.
- The chart identifies **KSC LC-39A** as SpaceX's most reliable launch site in this dashboard.



Payload Mass vs. Launch Success Across AllSites

- The scatter plot compares payload mass with launch outcome for all launch sites.
- Successes occur across the full payload range up to about **10,000 kg**.
- The **2,000–6,000 kg** range shows a dense cluster of successful launches.
- **FT** boosters show the highest concentration of success in the mid-range payload band.
- High-payload successes are mostly associated with **FT**, **B4**, and **B5** boosters.
- Older versions such as **v1.1** show more failures, especially at lower payload masses.

Correlation between Payload and Success for site KSC LC-39A



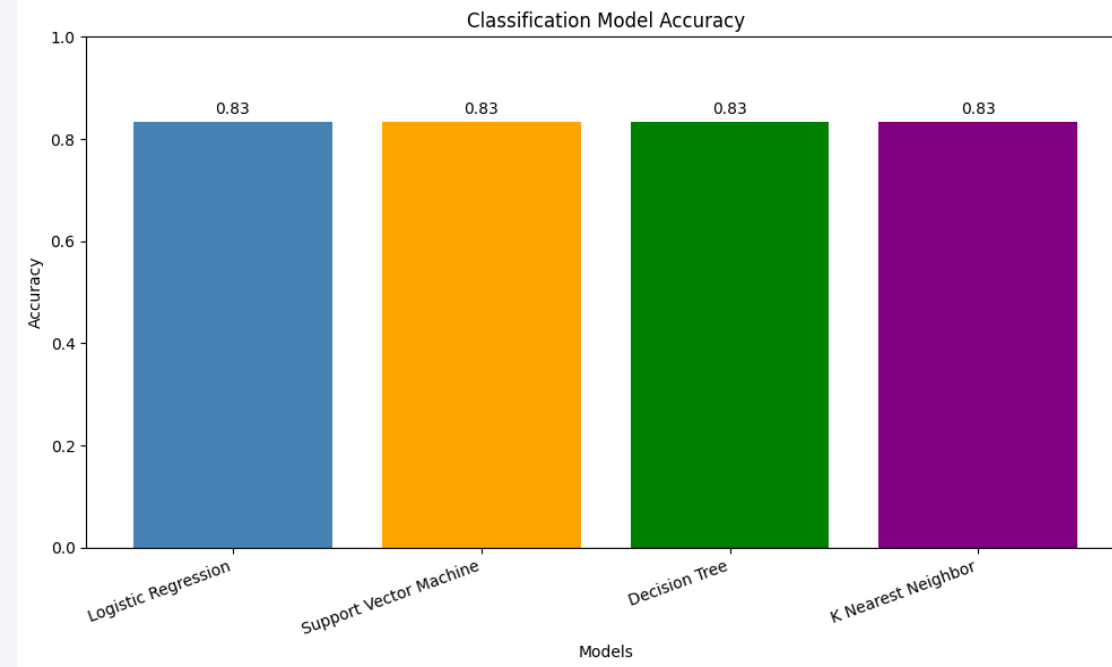


Section 5

Predictive Analysis (Classification)

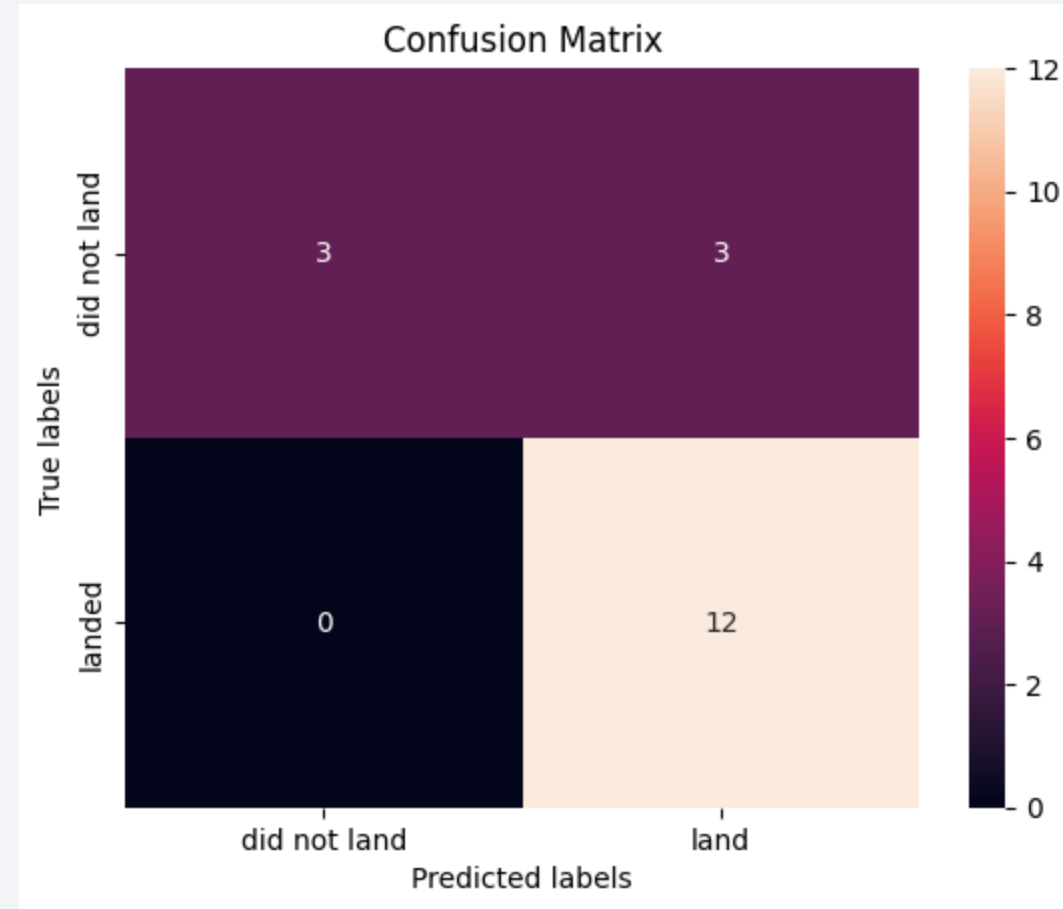
Classification Accuracy

- Visualize the built model accuracy for all built classification models, in a bar chart
- Find which model has the highest classification accuracy
- Logistic Regression, SVM, Decision Tree, and KNN each achieved 83.33% test accuracy.
- The bar chart shows a complete tie across all built classification models.
- The notebook identifies Logistic Regression as the best-performing method at 0.83 accuracy.
- Since all models perform the same, the dataset may have limited discriminatory power.
- Logistic Regression is a strong choice because it is simple, lightweight, and performs as well as the more complex models.



Confusion Matrix

- Logistic Regression achieved 83.33% test accuracy and was identified as the best method in the notebook.
- Diagonal cells show correct predictions; off-diagonal cells show misclassifications.
- True Positives (12): correctly predicted successful landings.
- True Negatives (3): correctly predicted failed landings.
- False Positives (3): predicted success, but the landing failed.
- False Negatives were not a major issue in this result.
- The model performs well overall but is slightly over-optimistic toward success predictions.



Conclusions

- Falcon 9 landing success can be predicted using historical launch data.
- EDA showed that flight number, launch site, payload, and orbit affect landing outcomes.
- Success rates improved over time, suggesting stronger operational experience.
- Classification models achieved about 83.33% accuracy on the test set.
- The project shows how data science can support reusable spaceflight decision-making.

Appendix

- Everything about this capstone project can be found in this GitHub repo

<https://github.com/omarfaruqe/DS-Capstone>

Thank you!

